

1. Let R be the initial radius of orbit and m be the mass of orbiting object. Then after the change in mass $M \rightarrow M'$, the perihelion (nearest point of orbit) is $R_p = R$ while the aphelion (furthest point) is R_a . Initial velocity is $V = \sqrt{\frac{GM}{R}}$.

By conservation of energy,

$$\frac{1}{2}mV^2 - \frac{GmM}{R} = \frac{1}{2}mV_a^2 - \frac{GmM'}{R_a}$$

and conservation of angular momentum,

$$mRV = mR_aV_a$$

putting everything together:

$$RV = R_a \sqrt{V^2 - 2GM' \left(\frac{1}{R} - \frac{1}{R_a} \right)}$$

Squaring both sides,

$$\begin{aligned} GMR &= R_a^2 \left(\frac{GM}{R} - 2GM' \left(\frac{1}{R} - \frac{1}{R_a} \right) \right) \therefore 1 = \left(\frac{R_a}{R} \right)^2 - \frac{2M'}{M} \left[\left(\frac{R_a}{R} \right)^2 - \frac{R_a}{R} \right] \\ &\therefore \left(1 - \frac{2M'}{M} \right) \left(\frac{R_a}{R} \right)^2 + \frac{2M'}{M} \left(\frac{R_a}{R} \right) - 1 = 0 \end{aligned}$$

This quadratic may initially seem daunting, but note $R_a = R$ is a root.

$$\left(\frac{R_a}{R} - 1 \right) \left[\left(1 - \frac{2M'}{M} \right) \frac{R_a}{R} + 1 \right] = 0 \therefore \frac{R_a}{R} = \frac{1}{\frac{2M'}{M} - 1}$$

Then we can find the eccentricity

$$e = \frac{R_a - R}{R_a + R} = \frac{M}{M'} - 1$$

Alternatively, we simply use the known result that total energy of a Kepler orbit $E = K + U = -\frac{GMm}{2a}$.

Initially,

$$E = K + U = \frac{GMm}{2R} - \frac{GMm}{R} = -\frac{GMm}{2R}$$

Finally (K is constant),

$$E = K + U = \frac{GMm}{2R} - \frac{GM'm}{R} = -\frac{GM'm}{2a}$$

This gives

$$\frac{R}{a} = \frac{2M' - M}{M'}$$

and eccentricity

$$e = \frac{a-R}{a} = 1 - \frac{R}{a} = \frac{M-M'}{M'} = \frac{M}{M'} - 1$$

2. (a) The path of the light ray is the straight line segment from A to B.

(b) (i) $l = \sqrt{(x - x_A)^2 + y_A^2} + \sqrt{(x_B - x)^2 + y_B^2}$

You know that it is reflection, and the angles should be the same. To rigorously show it, we find the local minimum of l (which corresponds to path of least time $\frac{nl}{c}$).

$$\frac{dl}{dx} = \frac{(x - x_A)}{\sqrt{(x - x_A)^2 + y_A^2}} - \frac{(x_B - x)}{\sqrt{(x_B - x)^2 + y_B^2}} = 0 \therefore \sin \theta_A - \sin \theta_B = 0$$

where θ_A is angle of incidence, θ_B is angle of reflection. Thus

$$\theta_A = \theta_B$$

(c) Let the light ray pass through the y-axis at point $(0, y)$. Then the optical path length is $l_n = n_1 \sqrt{x_A^2 + (y - y_A)^2} + n_2 \sqrt{x_B^2 + (y_B - y)^2}$.

Minimum time l_n/c corresponds to minimum l_n :

$$\frac{dl_n}{dy} = n_1 \frac{y - y_A}{\sqrt{x_A^2 + (y - y_A)^2}} - n_2 \frac{y_B - y}{\sqrt{x_B^2 + (y_B - y)^2}} = 0 \therefore n_1 \sin \theta_A - n_2 \sin \theta_B = 0$$

where θ_A is angle of incidence, θ_B is angle of reflection. Thus

$$n_1 \sin \theta_A = n_2 \sin \theta_B$$

which is exactly Snell's Law.

3. First thing to note is that the masses of each particle are the same, so $N \propto M$ (the number of particles is proportional to the mass of the particles). We work in ratios:

$$\begin{aligned} N_{Cl} : N_{Ar} : N_S &= 20 : 0.36 \cdot \frac{98.1}{1.9} : 0.36 \\ \therefore N_{Cl} : (N_{Ar} + N_S) &= 20 : 0.36 \cdot \frac{100}{1.9} = e^{-\lambda t} : (1 - e^{-\lambda t}) \\ \therefore e^{-\lambda t} : 1 &= N_{Cl} : N_{total} = 20 : \left(20 + 0.36 \cdot \frac{100}{1.9}\right) \end{aligned}$$

Thus

$$\begin{aligned} \lambda &= -\ln \left(\frac{20}{20 + 0.36 \cdot \frac{100}{1.9}} \right) \cdot \frac{1}{t} = \frac{\ln 2}{t_{1/2}} \therefore t_{1/2} = t \cdot \frac{\ln 2}{-\ln \left(\frac{20}{20 + 0.36 \cdot \frac{100}{1.9}} \right)} \\ &= \mathbf{301604 \text{ years} \approx 300000 \text{ years (2s.f.)} \end{aligned}$$

4. Use recursion to resolve the infinite capacitor circuit.

$$C_{eff} = C \parallel (C \oplus C_{eff} \oplus C) = C + \frac{1}{\frac{1}{C} + \frac{1}{C_{eff}} + \frac{1}{C}} = C + \frac{CC_{eff}}{2C_{eff} + C}$$

$$(2C_{eff} + C)(C_{eff} - C) = CC_{eff} \div 2C_{eff}^2 - 2CC_{eff} - C^2 = 0 \div C_{eff} = \left(\frac{1}{2} \pm \frac{\sqrt{3}}{2}\right)C$$

Reject the negative root, so $C_{eff} = \left(\frac{1}{2} + \frac{\sqrt{3}}{2}\right)C$

The AC source voltage is already given as V_{rms} and the current measures I_{rms} .

The impedance is $Z = \frac{1}{i\omega C_{eff}}$. So

$$I_{rms} \div \frac{V_{rms}}{|Z|} = V_{rms}\omega \left(\frac{1}{2} + \frac{\sqrt{3}}{2}\right)C = 220 \cdot 2\pi \cdot 50 \cdot \left(\frac{1}{2} + \frac{\sqrt{3}}{2}\right) \cdot 10^{-6} = \mathbf{0.0944 \text{ A}}$$

If you want the function, this is $I(t) = i\omega \left(\frac{1+\sqrt{3}}{2}C\right) (220\sqrt{2})e^{i\omega t}$, $\omega = 2\pi f$.

5. (a) If you already know the wave fourvector $\underline{k} = (k_x, k_y, k_z, i\frac{\omega}{c})$ then this problem is trivial (remember the position fourvector $\underline{r} = (x, y, z, ict)$, and that the dot product of two fourvectors is Lorentz-invariant). If you don't know this, then you have to figure out what how the question is guiding you to derive it.

The question gives you the fact that $k_x x + k_y y + k_z z - \omega t$ is Lorentz-invariant, i.e.

$$k'_x x' + k'_y y' + k'_z z' - \omega' t' = k_x x + k_y y + k_z z - \omega t$$

Simply substitute in your Lorentz transformation:

$$k'_x(\gamma x - \gamma\beta ct) + k'_y y + k'_z z - \omega' \left(\gamma t - \frac{\beta}{c} x\right) = k_x x + k_y y + k_z z - \omega t$$

Rearrange:

$$\left(\gamma k'_x + \frac{\beta}{c} \omega'\right)x + k'_y y + k'_z z - (\gamma \omega' + \gamma\beta c k'_x)t = k_x x + k_y y + k_z z - \omega t$$

This is true for all x, y, z, t . So we can compare coefficients

$$\begin{cases} \gamma k'_x + \frac{\gamma\beta}{c} \omega' = k_x \\ k'_y = k_y \\ k'_z = k_z \\ \gamma \omega' + \gamma\beta c k'_x = \omega \end{cases}$$

Rearranging:

$$\begin{cases} k'_x = \gamma k_x - \frac{\gamma\beta}{c} \omega \\ k'_y = k_y \\ k'_z = k_z \\ \omega' = \gamma \omega - \gamma\beta c k_x \end{cases}$$

Thus

$$c\mathbf{k}' = \begin{pmatrix} \gamma(ck_x - \beta\omega) \\ ck_y \\ ck_z \end{pmatrix}, \quad \omega' = \gamma\omega - \gamma\beta ck_x$$

(b) Incident photon has

$$c\mathbf{k}_0 = \begin{pmatrix} \omega_i \cos \theta_i \\ \omega_i \sin \theta_i \\ 0 \end{pmatrix}, \quad \omega_0 = \omega_i$$

Transforming into rest frame of mirror:

$$c\mathbf{k}'_0 = \begin{pmatrix} \gamma\omega_i \cos \theta_i - \gamma\beta\omega_i \\ \omega_i \sin \theta_i \\ 0 \end{pmatrix} = \omega_i \begin{pmatrix} \gamma \cos \theta_i - \gamma\beta \\ \sin \theta_i \\ 0 \end{pmatrix},$$

$$\omega'_0 = \gamma\omega_i - \gamma\beta\omega_i \cos \theta_i = (\gamma - \gamma\beta \cos \theta_i)\omega_i$$

Now reflection in rest frame of mirror is as expected:

$$c\mathbf{k}'_1 = \omega_i \begin{pmatrix} \gamma\beta - \gamma \cos \theta_i \\ \sin \theta_i \\ 0 \end{pmatrix}, \quad \omega'_1 = (\gamma - \gamma\beta \cos \theta_i)\omega_i$$

Transforming back to observer frame:

$$ck_1 = \omega_i \begin{pmatrix} \gamma(\gamma\beta - \gamma \cos \theta_i) + \gamma\beta(\gamma - \gamma\beta \cos \theta_i) \\ \sin \theta_i \\ 0 \end{pmatrix}$$

$$= \omega_i \begin{pmatrix} -\gamma^2(1 + \beta^2) \cos \theta_i + 2\gamma^2\beta \\ \sin \theta_i \\ 0 \end{pmatrix},$$

$$\omega_1 = [\gamma(\gamma - \gamma\beta \cos \theta_i) + \gamma\beta(\gamma\beta - \gamma \cos \theta_i)]\omega_i$$

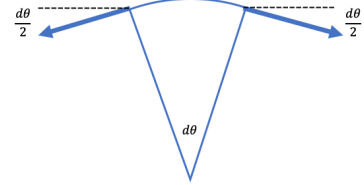
$$= (\gamma^2(1 + \beta^2) - 2\gamma^2\beta \cos \theta_i)\omega_i$$

Expressing in terms of β :

$$ck_1 = \omega_i \begin{pmatrix} -\frac{1 + \beta^2}{1 - \beta^2} \cos \theta_i + \frac{2\beta}{1 - \beta^2} \\ \sin \theta_i \\ 0 \end{pmatrix}, \quad \omega_1 = \left(\frac{1 + \beta^2}{1 - \beta^2} - \frac{2\beta}{1 - \beta^2} \cos \theta_i \right) \omega_i$$

(ii) energy $E = \hbar\omega_1 = \hbar \left(\frac{1 + \beta^2}{1 - \beta^2} - \frac{2\beta}{1 - \beta^2} \cos \theta_i \right) \omega_i$

6. (a) (i) Looking at the pressure equation, it is essentially $P = \frac{F}{A}$. Usually, surface tension $F = \sigma \times L$, $P = \sigma \left(\frac{1}{R_x} + \frac{1}{R_y} \right)$. This means that the provided F formula is the tension, not the coefficient of surface tension. However, it does not make much sense to define "tension" on a balloon, because the more rubber you consider, the more tension there is. Hence, the Force equation provided should be "tension per degree".



Consider a small square patch of sides $dl = r d\theta$.

$$dF = F_r \times d\theta = \frac{C_0(r \times d\theta)}{r_0^2} \left[1 - \left(\frac{r_0}{r} \right)^6 \right]$$

The normal component is $dF \cdot \sin \frac{d\theta}{2} \approx dF \cdot \frac{d\theta}{2}$, and there are 4 of them.

The four tensions are with constant C_0 . So the net force is in normal direction:

$$dF = 4 \frac{C_0 dl}{r_0^2} \left[1 - \left(\frac{r_0}{r} \right)^6 \right] \frac{d\theta}{2} = \frac{2C_0}{r_0^2 r} \left[1 - \left(\frac{r_0}{r} \right)^6 \right] dl^2 \therefore P = \frac{dF}{dl^2} = \frac{2C_0}{r_0^2 r} \left[1 - \left(\frac{r_0}{r} \right)^6 \right]$$

which is the desired relation, with proportionality constant $C = 2C_0$.

If you are wondering, you can take a circular piece of radius $\frac{dl}{2} = \frac{r d\theta}{2}$ too.

$$dF = F_r \times \frac{d\theta}{2} \times d\phi = \frac{C_0(r \times d\theta \times d\phi)}{2r_0^2} \left[1 - \left(\frac{r_0}{r} \right)^6 \right]$$

Overall normal component:

$$\int_0^{2\pi} \frac{C_0 dl}{2r_0^2} \left[1 - \left(\frac{r_0}{r} \right)^6 \right] \frac{d\theta}{2} d\phi = \frac{\pi C_0}{2r_0^2 r} \left[1 - \left(\frac{r_0}{r} \right)^6 \right] dl^2$$

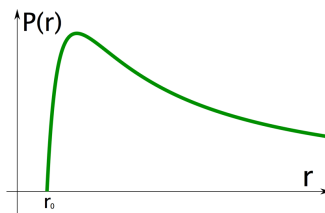
$$P = \frac{dF}{\pi \left(\frac{dl}{2} \right)^2} = \frac{2C_0}{r_0^2 r} \left[1 - \left(\frac{r_0}{r} \right)^6 \right]$$

(ii) Maximum pressure is a stationary point.

$$\frac{dP}{dr} = 0 = \frac{C}{r_0^2} \left\{ -\frac{1}{r^2} \left[1 - \left(\frac{r_0}{r} \right)^6 \right] - \frac{1}{r} \cdot 6 \left(\frac{r_0}{r} \right)^5 \cdot \frac{-r_0}{r^2} \right\} \therefore 1 - \left(\frac{r_0}{r} \right)^6 - 6 \left(\frac{r_0}{r} \right)^6 = 0$$

$$\therefore \left(\frac{r_0}{r} \right)^6 = \frac{1}{7} \therefore r = 7^{1/6} r_0$$

(iii) It looks like this:



(b) (i) The pressure of the gas in both balloons at equilibrium are the same.

This means that the equation

$$P = \frac{C}{r_0^2 r} \left[1 - \left(\frac{r_0}{r} \right)^6 \right]$$

has two roots $r = 6^{1/6} r_0$ and $r = r_{bf} = \kappa_f r_0$.

Equating:

$$\frac{1}{6^{1/6}} \left[1 - \frac{1}{6} \right] = \frac{1}{\kappa_f} \left[1 - \frac{1}{\kappa_f^6} \right]$$

Using iteration method

$$\kappa_f := \frac{6^{7/6}}{5} \left[1 - \frac{1}{\kappa_f^6} \right]$$

with initial value 1.5, we get $\kappa_f = 1.421766$ (to ensure that our value is accurate, note that starting value 1.4217655 causes increasing sequence while starting 1.4217665 causes decreasing sequence).

$$r_{bf} = \mathbf{1.42} r_0$$

Note: I prefer the newton-raphson method: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

[https://brilliant.org/wiki/newton-raphson-method/#:-:text=The%20Newton%20Raphson%20method%20\(also, straight%20line%20tangent%20to%20it](https://brilliant.org/wiki/newton-raphson-method/#:-:text=The%20Newton%20Raphson%20method%20(also, straight%20line%20tangent%20to%20it)

(ii) The gases obey $PV = NRT$, and total number of gas particles is conserved, so $P_1 V_1 + P_2 V_2$ is conserved, so $r_1^2 \left[1 - \left(\frac{r_0}{r_1} \right)^6 \right] + r_2^2 \left[1 - \left(\frac{r_0}{r_2} \right)^6 \right]$ is constant. Let $r_{bi} = \kappa_i r_0$

$$\begin{aligned} (7^{1/6})^2 \left[1 - \frac{1}{7} \right] + \kappa_i^2 \left[1 - \frac{1}{\kappa_i^6} \right] &= (6^{1/6})^2 \left[1 - \frac{1}{6} \right] + 1.421766^2 \left[1 - \frac{1}{1.421766^6} \right] \\ \kappa_i^2 \left[1 - \frac{1}{\kappa_i^6} \right] &= 1.651300 \end{aligned}$$

Using iteration method

$$\kappa_i := \sqrt{1.651300 + \frac{1}{\kappa_i^4}}$$

with initial value $\kappa_i = 1$ gives $\kappa_i = 1.386363$. So

$$r_{bi} = \mathbf{1.39} r_0$$

(c) This is a tall order, but let's try to form the integral.

$$t = \int dt = \int_{r_{bi}}^{r_{bf}} \frac{1}{\left(\frac{dr_b}{dt}\right)} dr_b = \int_{r_{bi}}^{r_{bf}} \frac{\frac{d}{dr_b}(P_b V_b)}{\frac{d}{dt}(P_b V_b)} dr_b$$

We chose to rewrite it in terms of derivatives $P_b V_b$ because: we know P_b and V_b in terms of r_b , so the numerator is settled; also, $P_b V_b = N_b kT \propto N_b$ so the time derivative is related to the flow rate of molecules from one balloon to the other, which we can probably work out in terms of the pressure difference and other parameters.

Firstly (*):

$$\begin{aligned} \frac{d}{dr_b}(P_b V_b) &= \frac{d}{dr_b} \left[\frac{C}{r_0^2 r_b} \left(1 - \left(\frac{r_0}{r_b} \right)^6 \right) \cdot \frac{4\pi}{3} r_b^3 \right] \\ &= \frac{4\pi}{3} \cdot \frac{C}{r_0^2} \cdot \left[2r_b \left(1 - \left(\frac{r_0}{r_b} \right)^6 \right) + r_b^2 6 \left(\frac{r_0}{r_b} \right)^5 \frac{r_0}{r_b^2} \right] = \frac{4\pi}{3} \cdot \frac{C}{r_0^2} \cdot \left[2r_b + 4 \frac{r_0^6}{r_b^5} \right] \end{aligned}$$

Next,

$$\frac{d}{dt}(P_b V_b) = kT \frac{dN_b}{dt}$$

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Consider a single balloon, losing air through a small hole on its surface of area  $A$ . The root mean square speed in the direction normal to the hole is  $v_x = \sqrt{\frac{kT}{m}}$ .

The volumetric flow rate out of the hole is thus

$$\frac{dV}{dt} = \frac{1}{2} A v_x$$

where the factor of half comes from the fact that root mean square speed consider particles moving in both directions.

The corresponding molar flow rate is

$$\frac{dN}{dt} = \frac{P}{kT} \cdot \frac{dV}{dt} = \frac{P}{kT} \cdot \frac{A}{2} \sqrt{\frac{kT}{m}}$$

For a more rigorous derivation, see below.

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Therefore the net flow rate of molecules is

$$\frac{dN_{net}}{dt} = \frac{P_s}{kT} \cdot \frac{A}{2} \sqrt{\frac{kT}{m}} - \frac{P_b}{kT} \cdot \frac{A}{2} \sqrt{\frac{kT}{m}} = \frac{(P_s - P_b)}{kT} \cdot \frac{A}{2} \sqrt{\frac{kT}{m}}$$

so (*)

$$\begin{aligned} \frac{d}{dt}(P_b V_b) &= kT \cdot \left[\frac{(P_a - P_b)}{kT} \cdot \frac{A}{2} \sqrt{\frac{kT}{m}} \right] = (P_a - P_b) \cdot \frac{A}{2} \sqrt{\frac{kT}{m}} \\ &= \frac{C}{r_0^2} \left[\frac{1}{r_s} \left(1 - \left(\frac{r_0}{r_s} \right)^6 \right) - \frac{1}{r_b} \left(1 - \left(\frac{r_0}{r_b} \right)^6 \right) \right] \frac{A}{2} \sqrt{\frac{kT}{m}} \end{aligned}$$

Putting the 2 (*)'ed equations into the original integral:

$$t = \int_{r_{bi}}^{r_{bf}} \frac{\frac{4\pi}{3} \cdot \frac{C}{r_0^2} \cdot \left[2r_b + 4 \frac{r_0^6}{r_b^5} \right]}{\frac{C}{r_0^2} \left[\frac{1}{r_s} \left(1 - \left(\frac{r_0}{r_s} \right)^6 \right) - \frac{1}{r_b} \left(1 - \left(\frac{r_0}{r_b} \right)^6 \right) \right] \frac{A}{2} \sqrt{\frac{kT}{m}}} dr_b$$

Taking out the factors:

$$t = \frac{16\pi}{3A} \sqrt{\frac{m}{kT}} \int_{r_{bi}}^{r_{bf}} \frac{r_b \left[1 + 2 \left(\frac{r_0}{r_b} \right)^6 \right]}{\frac{1}{r_s} \left[1 - \left(\frac{r_0}{r_s} \right)^6 \right] - \frac{1}{r_b} \left[1 - \left(\frac{r_0}{r_b} \right)^6 \right]} dr_b$$

Here is a more rigorous derivation of the effusion rate. The continuous probability distribution of x-velocities (where the x-direction is normal to the hole) is

$$p(v_x) = Ae^{-\frac{mv_x^2}{2kT}}$$

where $A = \sqrt{\frac{m}{2\pi kT}}$ (u can derive this by integrating $p(v_x)dv_x$ from $-\infty$ to ∞ , which must equal 1).

Consider a gas in a cuboid. The amount of collisions that one particle will have with a wall of area A (the last dimension is length l) in time Δt is

$$\frac{|v_x|\Delta t}{2l}$$

Thus the average amount of collisions with the wall of area A in time Δt is

$$N \int_{-\infty}^{\infty} p(v_x) \frac{|v_x|\Delta t}{2l} dv_x = \frac{N}{l} \sqrt{\frac{kT}{2\pi m}} \Delta t$$

But $Al = V = \frac{NkT}{P} \therefore \frac{N}{l} = \frac{PA}{kT}$ which gives

$$\frac{dN}{dt} = \frac{PA}{kT} \sqrt{\frac{kT}{2\pi m}}$$

which is similar to above, but with a different factor. In fact this is more accurate (a similar derivation is here: <http://www.chem.hope.edu/~polik/Chem345-1997/gaseffusion/GasEffusion.html>). This results in a different factor:

$$t = \frac{8\pi}{3A} \sqrt{\frac{2\pi m}{kT}} \int_{r_{bi}}^{r_{bf}} \frac{r_b \left[1 + 2 \left(\frac{r_0}{r_b} \right)^6 \right]}{\frac{1}{r_s} \left[1 - \left(\frac{r_0}{r_s} \right)^6 \right] - \frac{1}{r_b} \left[1 - \left(\frac{r_0}{r_b} \right)^6 \right]} dr_b$$

However, the original expression in the actual paper is not this one but the one provided above instead.

7. (a) Consider a Gaussian surface which is a prism of small cross-sectional area A and (small) thickness Δt , positioned at a point on the surface such that the one face of area A is within the surface and one face is outside (so the normal vectors are perpendicular).

Applying Gauss' Law (and taking limit $\Delta t \ll \sqrt{A}$ and $A \rightarrow 0$):

$$\oiint \vec{E} \cdot d\vec{A} = E_{outside}^\perp \cdot A - E_{inside}^\perp \cdot A = \frac{q_{enclosed}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0}$$

Therefore

$$E_{outside}^\perp - E_{inside}^\perp = \frac{\sigma}{\epsilon_0}$$

(b) Let the sphere have radius R . The electric field at radius r must be directed radially, due to symmetry. Consider the Gaussian spherical surface of radius r :

$$\oiint \vec{E} \cdot d\vec{A} = E_r 4\pi r^2 = \frac{q_{enclosed}}{\epsilon_0} = \begin{cases} \frac{\frac{4}{3}\pi r^3 \rho}{\epsilon_0}, & r \leq R \\ \frac{\frac{4}{3}\pi R^3 \rho}{\epsilon_0}, & r > R \end{cases}$$

Thus

$$\vec{E} = \begin{cases} \frac{\pi \rho}{3\epsilon_0} \vec{r}, & r \leq R \\ \frac{\pi R^3 \rho}{3\epsilon_0 r^3} \vec{r}, & r > R \end{cases}$$

$$E_{inside} = E_{inside}^\perp = \lim_{r \rightarrow R^-} \frac{\pi \rho r}{3\epsilon_0} = \frac{\pi \rho R}{3\epsilon_0}, \quad E_{outside} = E_{outside}^\perp = \lim_{r \rightarrow R^+} \frac{\pi \rho R^3}{3\epsilon_0 r^2} = \frac{\pi \rho R}{3\epsilon_0}$$

Since $E_{outside} = E_{inside} \therefore E_{outside}^\perp - E_{inside}^\perp = 0 = \frac{\sigma}{\epsilon_0} \therefore \sigma = 0$ everywhere on the surface of the sphere.

(iii) The charge will be uniformly distributed on the surface, with charge-per-area $\sigma = \frac{Q}{4\pi R^2}$. Now consider a small patch of area dA on the sphere. What is the force on the patch due to the rest of the sphere?

The electric field just above the patch is $\frac{\sigma}{\epsilon_0}$. One can see this as the superposition of the electric field E_{patch} caused by the patch, and the electric field E_{rest} caused by the rest of the sphere. But $E_{patch} = \frac{\sigma}{2\epsilon_0}$ is the electric field right above a uniform-charged plate. Therefore $E_{rest} = \frac{\sigma}{2\epsilon_0}$ and the small patch experiences force $dF = \sigma dA \cdot E_{rest} = \sigma dA \frac{\sigma}{2\epsilon_0} = \frac{\sigma^2}{2\epsilon_0} dA$ radially outward.

We can therefore integrate by rings on the sphere ($dA = 2\pi R \sin \theta \cdot R d\theta$):

$$F = \int_0^{\frac{\pi}{2}} \frac{\sigma^2}{2\epsilon_0} 2\pi R \sin \theta \cdot R d\theta \cdot \cos \theta = \frac{\pi R^2 \sigma^2}{2\epsilon_0} \int_0^{\frac{\pi}{2}} \sin 2\theta d\theta = \frac{\pi R^2 \sigma^2}{2\epsilon_0} \cdot 1 = \frac{Q^2}{32\pi\epsilon_0 R^2}$$